

EE 446 TinyML - Homework 2

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Problem 1 - Data Preprocessing

Summary: The preprocessing code dropped the columns land, urgent, numfailedlogins, and numoutboundcmds. It then converted all non-normal attack labels to attack and used LabelEncoder on protocoltype, service, flag, and attack. This produces a numeric feature table suitable for visualization, DNN training, quantization, and Arduino deployment.

Problem 2 - Dimensionality Reduction Visualizations

Summary: The test set was visualized in 2D using t-SNE, PCA, and RBF KernelPCA. The plots use red points for attack traffic and blue points for normal traffic, as requested.

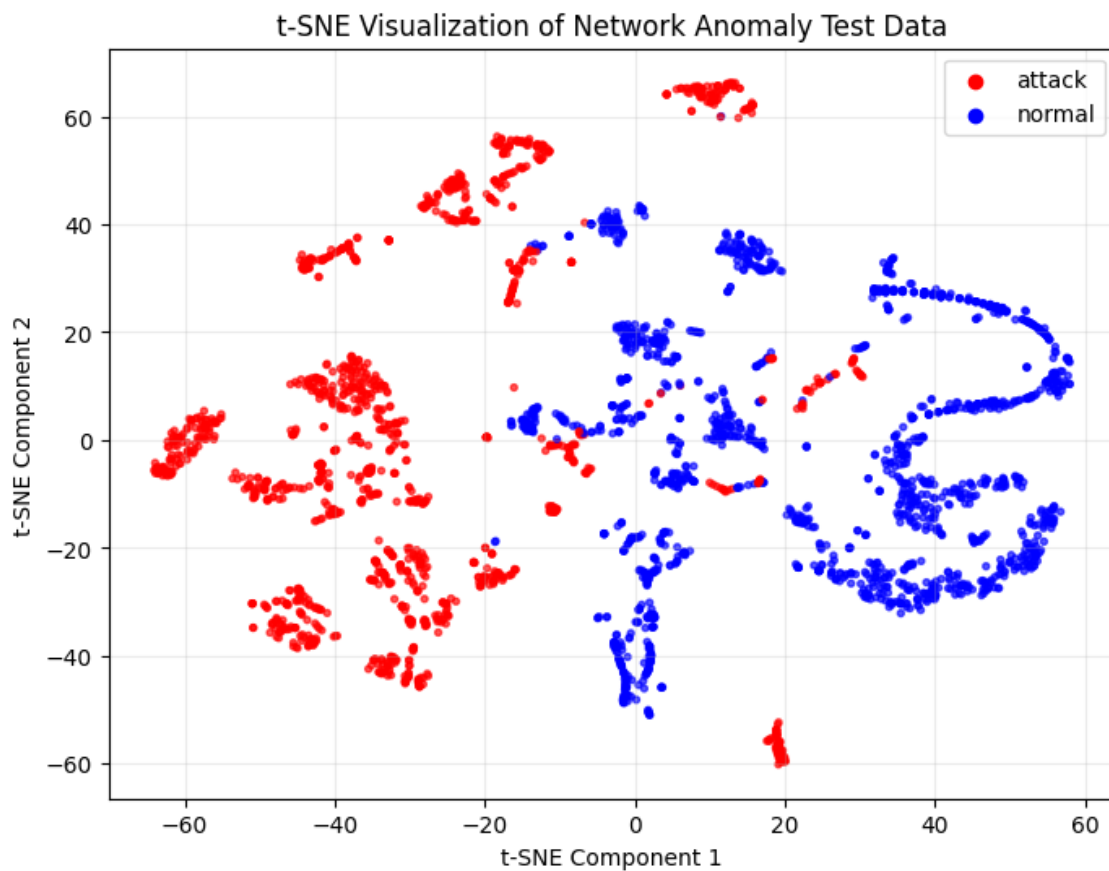


Figure 1. t-SNE visualization of the network anomaly test data.

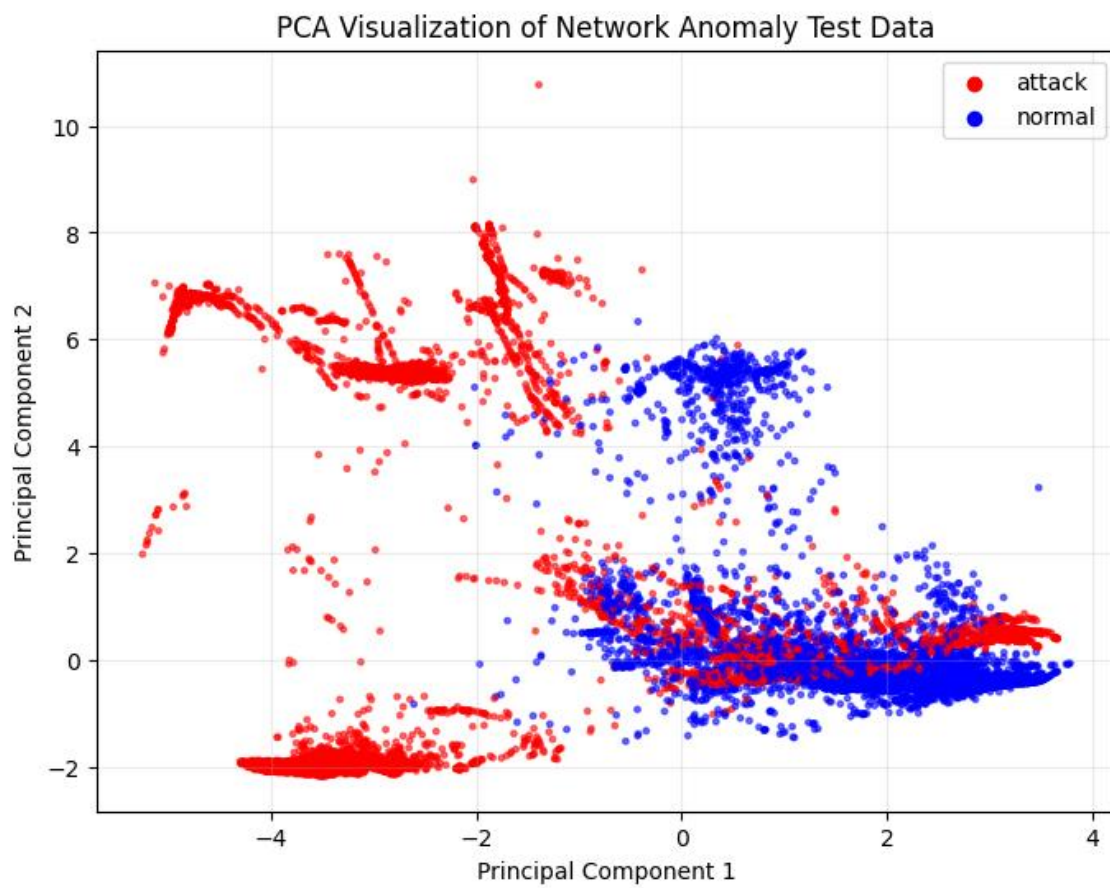


Figure 2. PCA visualization of the network anomaly test data.

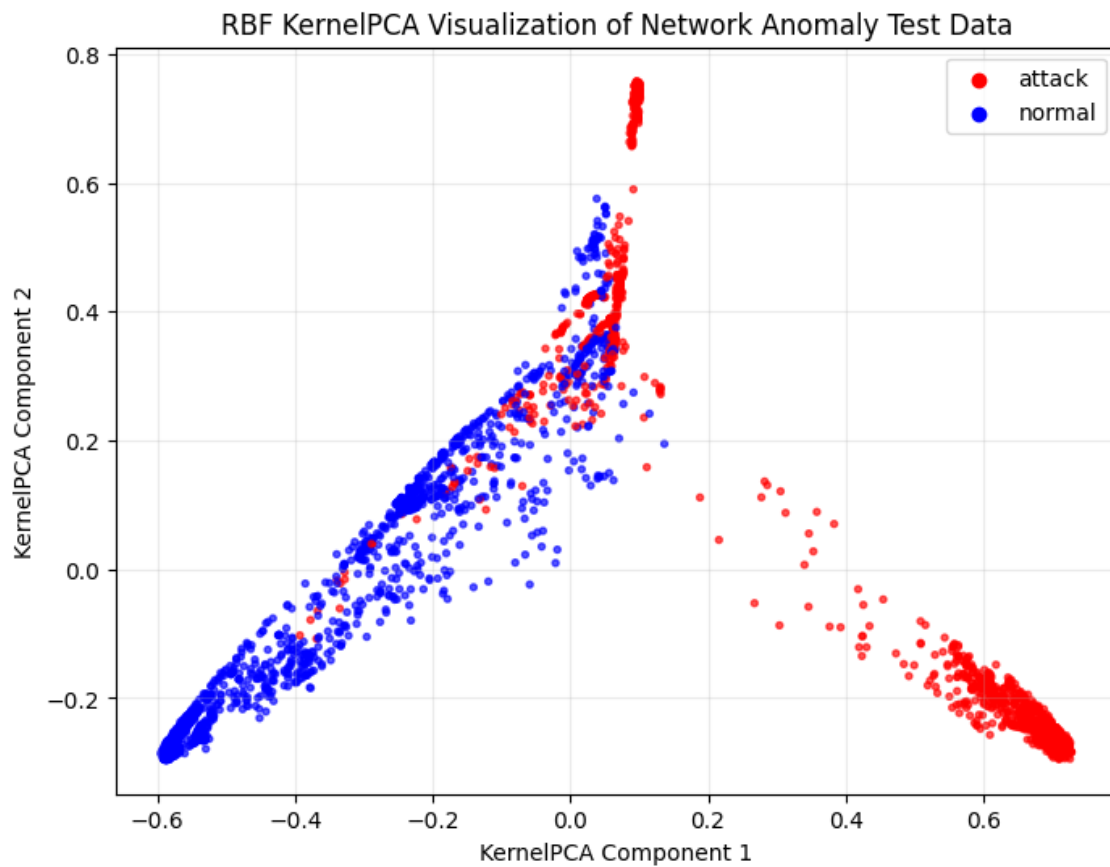


Figure 3. RBF KernelPCA visualization of the network anomaly test data.

Problem 3(c) - Base DNN Evaluation on the Test Set

Result: The base DNN uses a two-neuron softmax output layer and was evaluated on X_{test} using a classification report and confusion matrix. The test accuracy was approximately 0.99865.

Base model classification report:

	precision	recall	f1-score	support
attack	1.00	1.00	1.00	11784
normal	1.00	1.00	1.00	13411
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

Base model confusion matrix:

```
[[11773  11]
 [  23 13388]]
```

Base model test accuracy: 0.9986505258979956

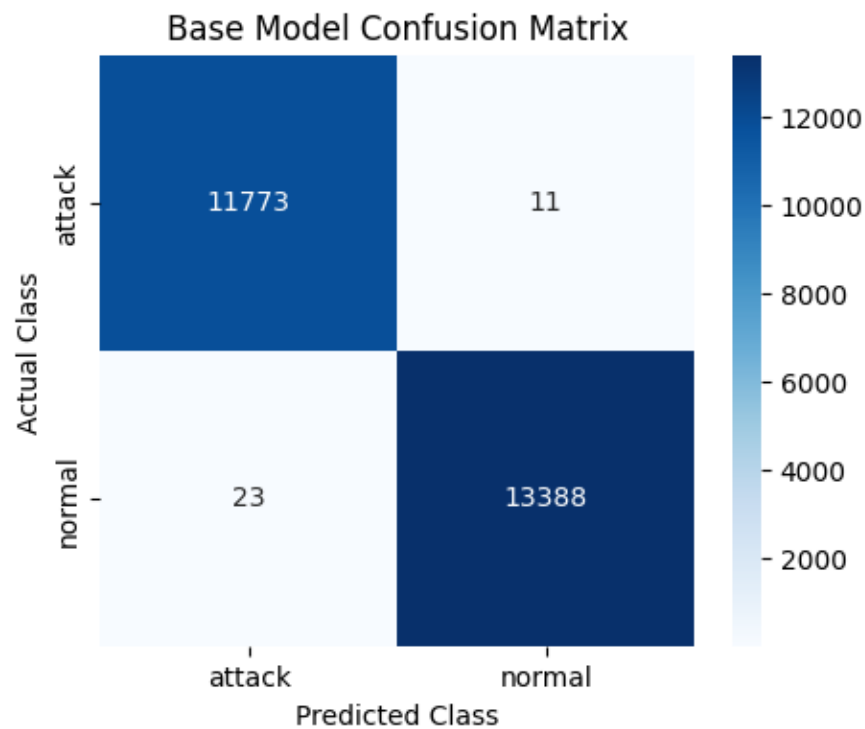


Figure 4. Base DNN confusion matrix.

Problem 4(b) - Dynamic Range Quantized TFLite Model Evaluation

Result: The base model was converted using dynamic range quantization and evaluated on X_{test} . Its measured test performance matched the base model on this run, with a test accuracy of approximately 0.99865.

Dynamic range quantized TFLite model classification report:

	precision	recall	f1-score	support
attack	1.00	1.00	1.00	11784
normal	1.00	1.00	1.00	13411
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

Dynamic range quantized TFLite model confusion matrix:

```
[[11773  11]
 [  23 13388]]
```

Quantized TFLite model test accuracy: 0.9986505258979956

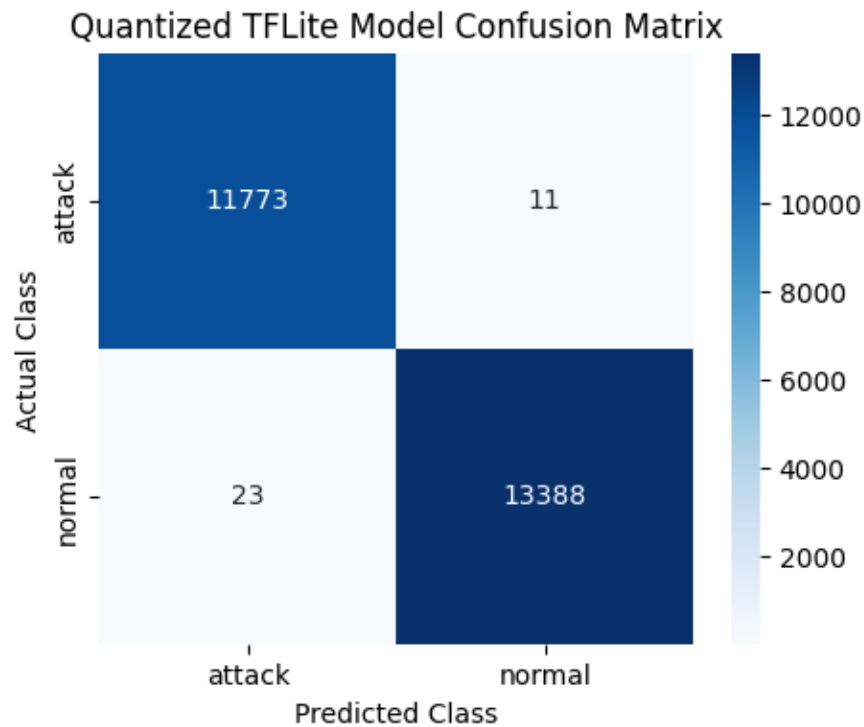


Figure 5. Dynamic range quantized TFLite model confusion matrix.

Problem 5 - Arduino Nano 33 BLE Sense Deployment

Deployment notes: The Arduino sketch was modified to load `network_model.h`, obtain the output tensor scores, choose the predicted class from the larger output value, and print the sample number, predicted class, actual class, and output scores to the Serial Monitor. A fully int8 model header was used for Arduino compatibility.

Problem 5(c) - First Five Serial Monitor Samples

Observation: The first five uploaded samples all matched the actual labels in the Serial Monitor output, giving 5/5 correct predictions for this deployment check.

```
Starting HW2 network anomaly inference sketch...
Model initialized successfully.
Input type: 9
Output type: 9
Input bytes: 38
Output bytes: 2
Sample #1 | Predicted Class: 1 | Actual Class: 1 | Output scores: [0.066406, 0.933594]
Sample #2 | Predicted Class: 1 | Actual Class: 1 | Output scores: [0.000000, 0.996094]
Sample #3 | Predicted Class: 0 | Actual Class: 0 | Output scores: [0.996094, 0.000000]
Sample #4 | Predicted Class: 0 | Actual Class: 0 | Output scores: [0.996094, 0.000000]
Sample #5 | Predicted Class: 1 | Actual Class: 1 | Output scores: [0.000000, 0.996094]
```

Figure 6. Serial Monitor output for the first five samples.

Problem 5(d) - Ten Additional Serial Monitor Samples

Observation: The second Arduino run used ten additional samples from the test set, excluding the first five. In the displayed output, Sample #1 was predicted as class 0 while its actual class was 1; the other nine samples matched. This gives 9/10 correct predictions for this small deployment check.

```
Starting HW2 network anomaly inference sketch...
Model initialized successfully.
Input type: 9
Output type: 9
Input bytes: 38
Output bytes: 2
Sample #1 | Predicted Class: 0 | Actual Class: 1 | Output scores: [0.996094, 0.000000]
Sample #2 | Predicted Class: 1 | Actual Class: 1 | Output scores: [0.000000, 0.996094]
Sample #3 | Predicted Class: 0 | Actual Class: 0 | Output scores: [0.996094, 0.000000]
Sample #4 | Predicted Class: 0 | Actual Class: 0 | Output scores: [0.996094, 0.000000]
Sample #5 | Predicted Class: 0 | Actual Class: 0 | Output scores: [0.996094, 0.000000]
Sample #6 | Predicted Class: 0 | Actual Class: 0 | Output scores: [0.996094, 0.000000]
Sample #7 | Predicted Class: 0 | Actual Class: 0 | Output scores: [0.996094, 0.000000]
Sample #8 | Predicted Class: 0 | Actual Class: 0 | Output scores: [0.996094, 0.000000]
Sample #9 | Predicted Class: 1 | Actual Class: 1 | Output scores: [0.000000, 0.996094]
Sample #10 | Predicted Class: 0 | Actual Class: 0 | Output scores: [0.996094, 0.000000]
```

Figure 7. Serial Monitor output for the ten additional samples.